

Route2Zero

Six-Month Pilot Plan · Metro Manila e-jeepney corridor prioritisation

SCREEN → VALIDATE → CO-DESIGN → ACT

A formal plan to convert a transparent screening MVP into a city-owned corridor shortlist for deeper electrification feasibility work.

Prepared by	Version	Date
Route2Zero team	2026 dashboard edition	17 August 2026

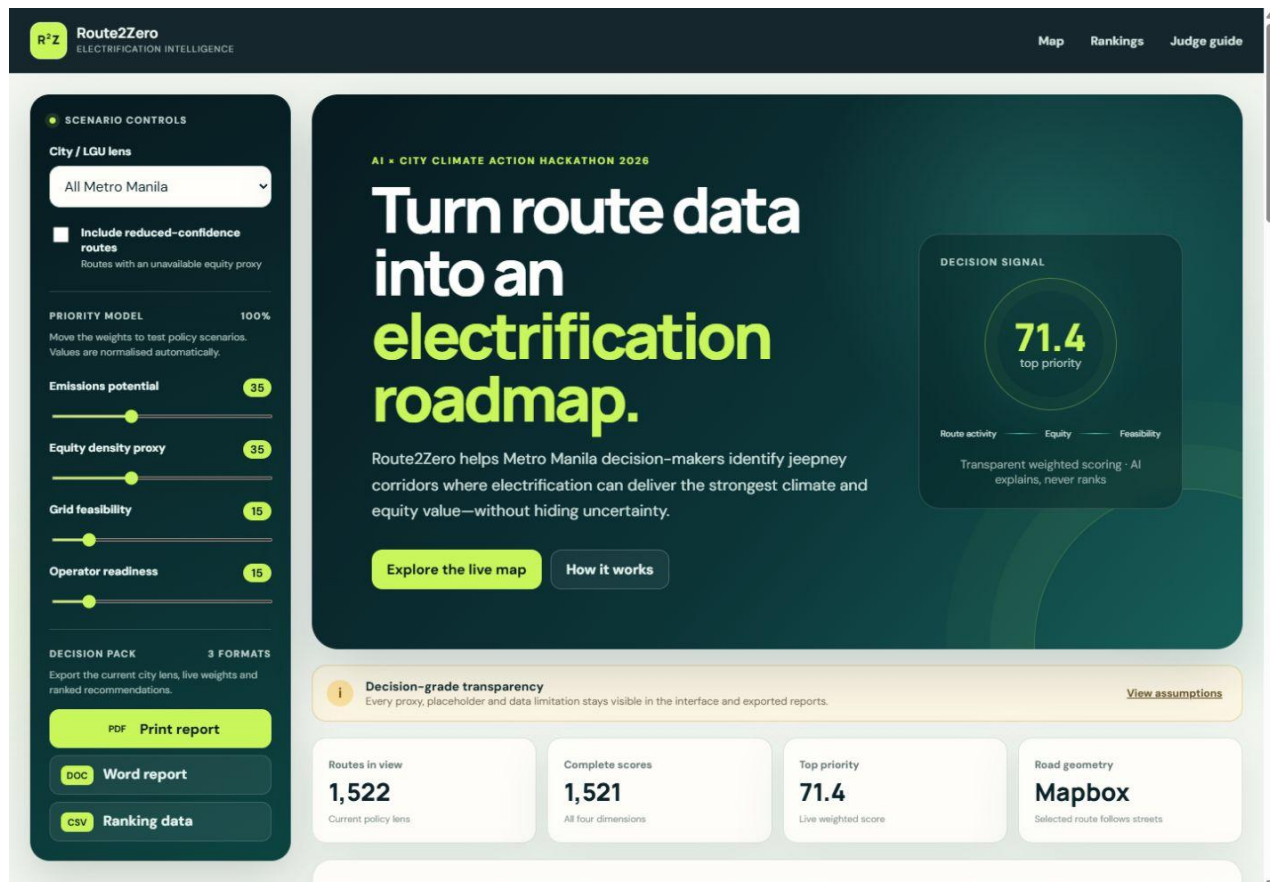


Figure 1. Live Route2Zero dashboard, 2026 edition.

A pilot built around decision quality

Route2Zero screens 1,522 Metro Manila jeepney route-direction records using four visible dimensions: emissions potential, equity density, grid feasibility and operator readiness. The default 35/35/15/15 weighting is adjustable and deterministic; optional AI explanations are downstream of the rank.

The pilot does not “deploy the ranking.” It validates the evidence behind a shortlist with cities, operators, drivers, riders, utilities and financing partners.

Current evidence	MVP status	Pilot response
Route universe	1,522 records	Confirm current operational relevance
Score completeness	1,521 complete	Resolve missing proxy and verify samples
Road geometry	Mapbox-selected route follows streets	Ground-check shortlisted corridors
Operator readiness	Neutral 50 placeholder	Replace with cooperative evidence

End-of-pilot outcome

- One transparent, city-validated corridor shortlist.
- Evidence cards for every shortlisted route.
- An exportable decision pack with assumptions, sources and unresolved risks.
- A repeatable governance process for updating the shortlist.

Prioritisation is a coordination problem

Electrification decisions span transport operations, climate value, affordability, grid readiness and the lived experience of operators and riders. These inputs are typically held by different institutions, measured at different scales and updated at different times.

Pilot thesis

A transparent screening model can accelerate coordination when it makes assumptions visible, supports scenario testing and converts uncertainty into specific evidence requests.

Planning failure mode	Consequence	Route2Zero response
Activity-only ranking	Ignores distributional value	Separate emissions and equity signals
Opaque composite score	Trade-offs cannot be challenged	Visible weights and components
Historic data treated as current	False operational confidence	Historic-data warning and validation gate
Readiness guessed silently	Weak implementation case	Explicit neutral placeholder

Design principle: no hidden certainty.

03 / OBJECTIVES

Five pilot objectives

- Validate whether shortlisted records correspond to current, meaningful operating corridors.
- Replace coarse or placeholder inputs with locally owned evidence where feasible.
- Co-design the policy lens and evidence thresholds with participating LGUs and operators.
- Test the dashboard and report exports in real decision meetings.
- Produce a replicable, accountable protocol for future corridor refreshes.

Decision questions

Question	Evidence needed	Owner
Is the corridor active and represented correctly?	Route ground-check + operator confirmation	Mobility workstream
Who gains and who bears transition costs?	Rider/operator engagement + disaggregated indicators	Equity workstream
Can charging be staged safely?	Depot, feeder, tariff and schedule screen	Grid/finance workstream
Is the shortlist accepted as a planning input?	Recorded decision session	Pilot lead

From transparent evidence to accountable action

Input	Activity	Output	Near-term outcome
Audited GTFS + proxies	Screen and rank	Challengeable shortlist	Faster evidence targeting
LGU and operator knowledge	Validate and annotate	Evidence cards	Shared understanding
Grid + finance context	Feasibility screen	Staging options	Reduced implementation risk
Open reporting	Export decision pack	Traceable record	Accountable next step

If evidence is visible and jointly validated, the shortlist becomes a shared planning instrument—not a black-box recommendation.

Assumptions to test

- Participating stakeholders have sufficient mandate and time.
- A corridor-level unit of analysis is useful for early screening.
- Proxy transparency improves trust more than false precision.
- Evidence gaps can be prioritised within six months.

What is inside—and outside—the pilot

In scope	Out of scope
Route screening and scenario analysis	Vehicle procurement recommendation
Selected route road-path visualisation	Engineering design of charging sites
Operator/rider validation workflow	Credit underwriting or fare policy
Grid and finance pre-feasibility screen	Real-time fleet dispatch
PDF, Word and CSV decision exports	Claim of current-service authority

Geographic focus

Select three participating LGUs using readiness, data access and stakeholder availability—not model rank alone. Cross-boundary corridors remain visible because operational and rider impacts do not stop at administrative borders.

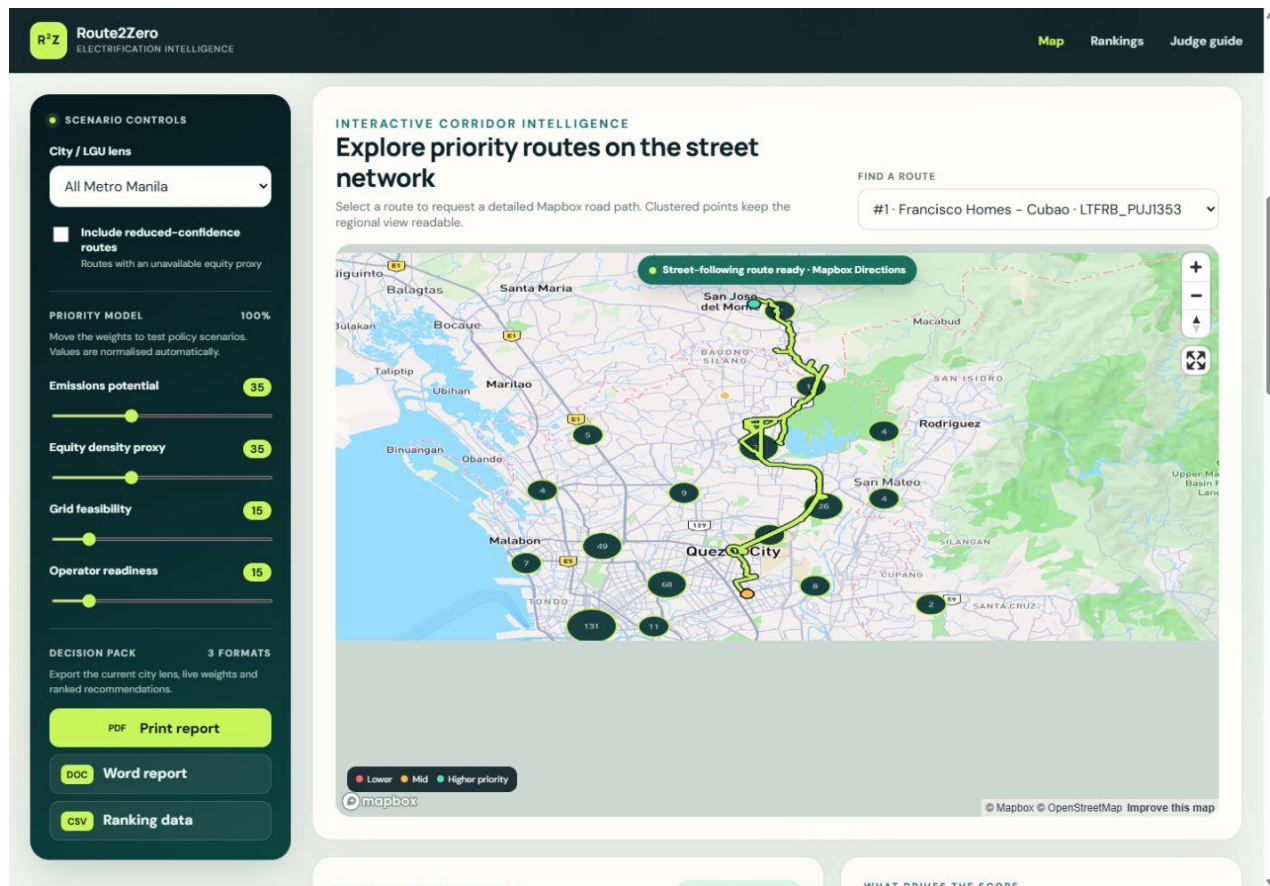


Figure 2. Selected route rendered on a Mapbox Directions street path.

Data refresh and provenance

Establish a dated, reproducible evidence baseline and document the gap between historic GTFS records and current operations.

Task	Method	Deliverable
Source register	Record licence, date, scale and owner	Versioned provenance log
GTFS re-audit	Integrity, calendar and route universe checks	Updated audit report
Proxy review	Validate units, ranges and confidence fields	Data dictionary
Change control	Record transforms and approvals	Refresh protocol

Acceptance criteria

- Every dashboard field maps to a named source or labelled placeholder.
- Historic boundaries are visible in the interface and exports.
- No personally identifiable information is collected.
- Rebuild produces the same ranking from the same inputs.

Gate 1 input: approved evidence-acquisition plan.

Route validation and geometry

The selected route follows roads using Mapbox Directions, but routing output remains a visualisation aid. It does not replace operational validation.

Validation layer	Check	Evidence
Identity	Name, direction, terminals	Operator + LGU confirmation
Path	Street sequence and deviations	Field trace / GPS sample
Service	Hours, headway, vehicles	Operator log or structured interview
Constraints	Flooding, width, enforcement, turning	Joint field note

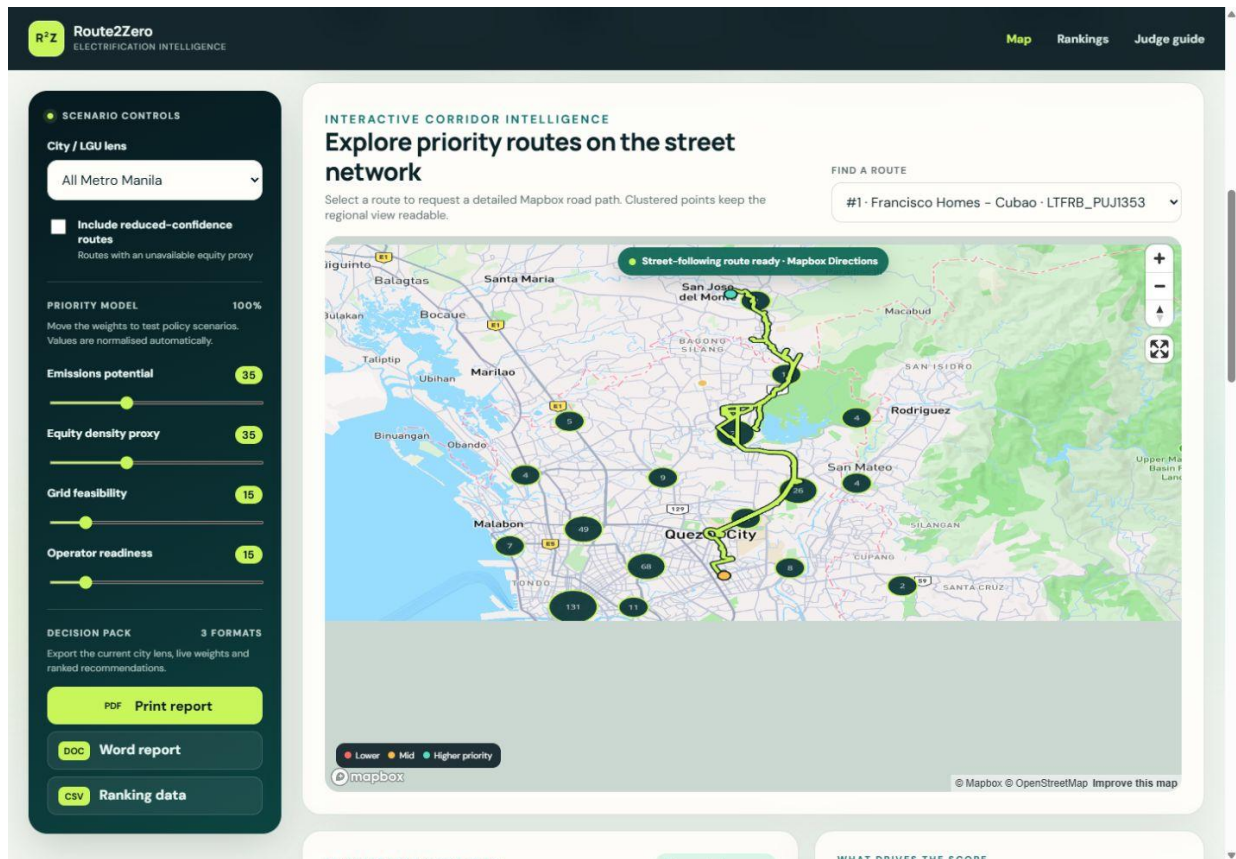


Figure 3. Street-following selected route with clustered regional records.

Output

A signed route-validation sheet for each shortlisted corridor, with unresolved differences retained rather than averaged away.

Operator and rider co-design

Readiness cannot be inferred from route activity. The pilot replaces the neutral operator placeholder with structured, consent-based evidence.

Participant	Key question	Safeguard
Cooperative leadership	What operating and finance conditions are feasible?	No commitment implied
Drivers	How would charging and vehicle finance affect work?	Anonymous synthesis
Riders	What access, reliability and fare risks matter?	Accessible sessions
LGU	Which constraints can the city influence?	Documented mandate

Minimum engagement standard

- At least one operator workshop per focus LGU.
- Separate driver and rider listening opportunities.
- Feedback log showing accepted, rejected and pending changes.
- No readiness score published without an evidence note.

Grid and finance pre-feasibility

The current grid score is a Luzon-level renewable-share proxy. The pilot treats it as context, then screens route-specific charging and affordability conditions.

Domain	Screening question	Pilot output
Depot / terminal	Where could vehicles dwell safely?	Candidate charging zones
Distribution grid	Is feeder capacity likely adequate?	Utility evidence request
Duty cycle	When and how long can vehicles charge?	Indicative charging windows
Finance	What cash-flow risks face operators?	Affordability questions
Energy mix	How should emissions claims be bounded?	Transparent calculation note

Pre-feasibility only: no engineering, tariff or lending decision is made in the MVP.

Dashboard, exports and decision sessions

The refreshed interface combines scenario controls, a Mapbox route view, ranked recommendations and judge-facing explanation sections. Every live scenario can be exported.

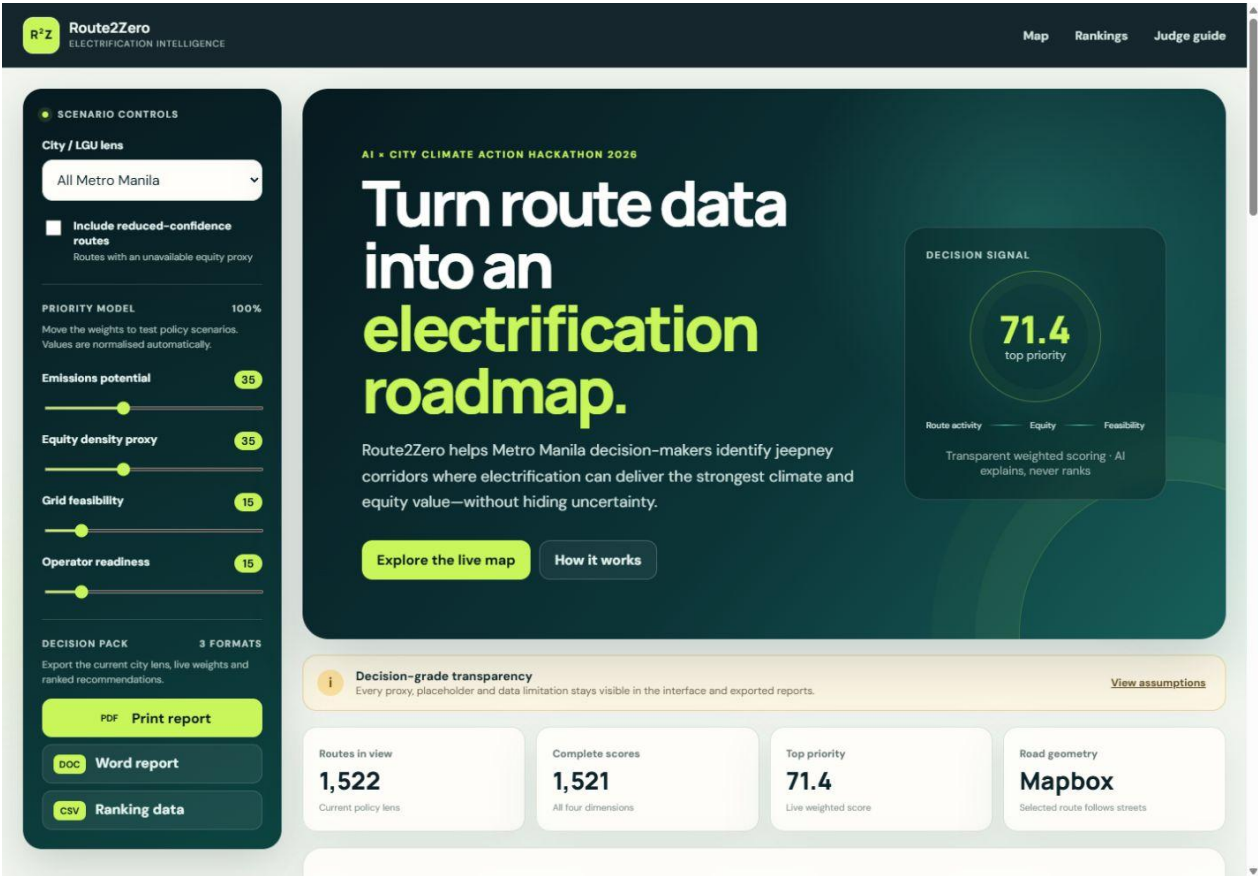


Figure 4. Desktop decision cockpit with live policy controls and report exports.

Format	Use	Required content
PDF	Formal circulation	Scenario, shortlist, assumptions
Word	Editable briefing	Narrative and evidence requests
CSV	Analyst handoff	Ranked records and active weights

Decision-session test

A city participant must be able to reproduce the active lens, explain why the top route appears, and identify at least one unresolved evidence gap.

Six months from screening to accepted decision pack

Month	Primary activity	Milestone outcome
1	Kickoff, source register, data audit	Scope and governance confirmed
2	Evidence plan and engagement design	Gate 1: evidence plan approved
3	Field validation and operator interviews	Initial evidence cards
4	Grid/finance screen and route reconciliation	Gate 2: routes validated
5	Scenario co-design and usability testing	Draft city shortlist
6	Decision sessions and handover	Gate 3: decision pack accepted

Cadence

- Weekly delivery stand-up.
- Fortnightly evidence review.
- Monthly steering meeting.
- Formal gate decision at months 2, 4 and 6.

12 / MILESTONE GATES

Evidence gates prevent premature recommendations

Gate	Decision	Minimum evidence	If not met
1 · End M2	Proceed to fieldwork	Scope, ethics, route sample and requests approved	Revise plan
2 · End M4	Proceed to shortlist co-design	Identity/path validation and stakeholder notes complete	Reduce or replace route
3 · End M6	Accept decision pack	Owners, confidence, next steps and dissent recorded	Issue qualified draft

A lower-confidence route may remain visible, but it cannot silently become decision-ready.

Gate authority

The steering group records each decision, rationale, conditions and dissent. Route2Zero team members prepare evidence but do not self-approve the city shortlist.

Named owners across product, data, mobility and engagement

Team member	Proposed pilot role	Primary accountability
John Marwin Ebona	Product and pilot lead	Decision process, delivery and handover
Prince Marl	Data and pipeline lead	Reproducibility and provenance
Joaquin Sarmiento	Mobility validation lead	Route sampling and operational checks
Isaac Marcus	Engineering lead	Dashboard, exports and deployment
Andrei Dela Cruz	Policy and governance lead	City process and decision records
Russel Mendez	Research and evidence lead	Methods and confidence notes
Tj Moreno	Engagement lead	Operator, driver and rider sessions
JM Palaganas	QA and documentation lead	Testing, accessibility and artifacts

Role assignments are proposed for the pilot and should be confirmed at kickoff alongside availability and relevant background evidence.

Every critical output has a single accountable owner

Output	Accountable	Contributors	Approver
Evidence register	Data lead	Research + mobility	Pilot lead
Validated route cards	Mobility lead	Engagement + LGU	Steering group
Dashboard release	Engineering lead	Product + QA	Pilot lead
Grid/finance screen	Policy lead	Utility/finance partners	Steering group
Decision pack	Product lead	All workstreams	City sponsor

Operating norms

- One owner per decision.
- Written evidence before score changes.
- Dissent and unresolved risks stay visible.
- No external claim is published without source and confidence review.

A lightweight structure with real decision rights

Forum	Frequency	Purpose	Decision right
Delivery stand-up	Weekly	Dependencies and blockers	Work allocation
Evidence review	Fortnightly	Data, methods and confidence	Accept evidence into model
Steering group	Monthly	Scope, risk and stakeholder alignment	Approve gate conditions
City decision session	M5–M6	Challenge and accept shortlist	Accept planning output

Change control

Any change to weights, source fields, confidence labels or shortlist status is recorded with author, date, reason, evidence and approver. The exported report states the active scenario.

Governance is part of the product—not a slide after the model.

Design the pilot with the people who operate and use the corridors

Phase	Stakeholders	Method	Output
Discover	LGUs, operators, utilities	Interviews + evidence mapping	Validated questions
Validate	Drivers, dispatchers, field staff	Walkthrough + route trace	Operational corrections
Co-design	Riders, operators, LGUs	Scenario workshop	Agreed policy lens
Decide	City sponsor + steering group	Recorded review	Accepted shortlist

Participation safeguards

- Plain-language purpose and limits.
- Consent for notes and recording.
- No collection of unnecessary personal data.
- Accessible scheduling and venue.
- Feedback loop showing how input changed the output.

Measure evidence quality, usability and adoption

Dimension	Indicator	Target	Source
Coverage	Shortlisted routes with complete evidence cards	100%	Evidence register
Validation	Priority records ground-checked	≥80%	Field log
Participation	Focus LGUs completing decision session	3	Meeting record
Usability	Users able to explain top rank and caveat	≥80%	Task test
Governance	Score changes with approval record	100%	Change log
Adoption	City sponsor accepts decision pack	1 accepted pack	Decision note

Targets are pilot management commitments, not claims about achieved impact. Climate and livelihood outcomes require follow-on implementation and measurement.

Treat the biggest credibility risks as delivery work

Risk	Likelihood	Impact	Mitigation	Owner
Historic GTFS misread as current	High	High	Persistent warning + current validation	Data
Road path mistaken for operational truth	Med	High	Ground-check and operator sign-off	Mobility
Operator concerns underrepresented	Med	High	Separate sessions and dissent log	Engagement
Grid proxy overinterpreted	High	Med	Utility evidence request + qualified language	Policy
Scenario weights politicised	Med	Med	Recorded workshop and versioned presets	Product
Deployment/configuration failure	Low	Med	Fallback public Mapbox token + smoke test	Engineering

Deterministic ranking, optional explanation, explicit boundaries

Control	Implementation	Verification
Ranking separation	AI cannot write score or rank fields	Pipeline tests
Provenance	Source and confidence columns retained	Data audit
Placeholder discipline	Operator 50 is labelled, not inferred	UI and export review
Privacy minimisation	No personal data required for scoring	Collection checklist
Human decision	City steering group accepts shortlist	Decision record
Model fallback	Deterministic explanation available	Function smoke test

Responsible-use statement

Route2Zero is suitable for screening and evidence prioritisation. It is not suitable as the sole basis for procurement, service cancellation, financing, fare changes or employment decisions.

Leave behind a maintainable process, not a one-off demo

Capability	Pilot handover	Scale pathway
Data refresh	Runbook + source register	Automated scheduled refresh after source agreements
Scoring	Versioned policy lens	City-specific presets with governance
Validation	Route evidence-card template	Operator-owned update cycle
Deployment	Netlify-ready static site	Shared domain and monitored deploy
Reporting	PDF/Word/CSV exports	Portfolio and investment-stage packs

Sustainability conditions

- Named institutional owner.
- Budget for data refresh and engagement.
- Clear escalation path for contested evidence.
- Annual review of proxies, weights and decision use.

A complete decision package

Deliverable	Due	Acceptance criterion
Provenance register + data dictionary	M2	Sources, dates, licences and confidence documented
Route validation set	M4	Identity, path, service and constraints reviewed
Operator/rider engagement synthesis	M4	Themes, dissent and actions logged
Grid/finance pre-feasibility note	M5	Questions and candidate zones documented
Refreshed dashboard and exports	M5	Desktop/mobile, Mapbox and reports pass QA
City-owned decision pack	M6	Shortlist, evidence, owners and next actions accepted

Definition of done: another authorised analyst can reproduce the pack and explain every unresolved limitation.

22 / RESOURCE ASSUMPTIONS

Plan by capability; price only with partners

The hackathon MVP does not fabricate a budget. The pilot should cost resources after participating LGUs, field scope, travel, workshop format, utility access and cloud/service terms are confirmed.

Cost band	Contents	Sizing driver
People	Product, data, mobility, policy, engagement, QA	Level of effort and partner contribution
Field validation	Travel, route traces, facilitation, accessibility	Routes and geographic spread
Stakeholder sessions	Venues, materials, participant support	Number and format of sessions
Technology	Hosting, map usage, document/report tooling	Traffic and Mapbox tier
Contingency	Weather, access and rescheduling	Fieldwork risk

Budget gate

A detailed costed work plan is produced after Month 1 scoping and approved before field commitments. All third-party service limits and public-token exposure are reviewed.

A credible roadmap starts with visible uncertainty

Route2Zero already demonstrates the end-to-end decision experience: adjustable policy weights, a ranked action queue, a street-following Mapbox route, responsive mobile access and downloadable reports. The six-month pilot turns that interface into a city-owned evidence and governance process.

SCREEN → VALIDATE → CO-DESIGN → ACT

Pilot ask	Commitment
Three partner LGUs	Decision owners and access to operational evidence
Operators and riders	Structured, consent-based validation
Utility and finance partners	Pre-feasibility questions and evidence
Route2Zero team	Transparent methods, responsive product and formal handover

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